

Data Collection and Cleaning in the Apparel and Textiles Industry

- **Project Title:** Data Collection and Cleaning in the Apparel and Textiles Industry
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1. Introduction

Data collection and cleaning are essential steps in every data analytics project. High-quality data improves the reliability of insights and supports better decision-making. In the apparel and textiles industry, clean and organized data helps businesses understand customer preferences, product trends, seasonal demand, and market performance. This project simulates a real-world data preparation workflow using publicly available datasets and Python.

2. Objective

The objective of this project is to identify publicly available datasets related to the apparel and textiles industry, collect relevant data, clean and preprocess the datasets using Python, and prepare an analysis-ready dataset for future business insights.

3. Data Source Identification

Dataset 1

Name: Dress Recommendation Dataset

Source: OpenML

Format: ARFF

Purpose:

- Consumer preferences
- Dress styles
- Product attributes
- Customer recommendations

Reason for Selection:

This dataset contains attributes such as style, price category, rating, season, material, decoration, and recommendation, making it suitable for understanding consumer preferences in the apparel industry.

Screenshot

	Style	Price	Rating	Size	Season	NeckLine	SleeveLength	WaistLine	Material	FabricType	Decoration	PatternType	Recommendation
0	Sexy	Low	4.6	M	Summer	o-neck	sleeveless	empire	?	chiffon	ruffles	animal	2
1	Casual	Low	0.0	L	Summer	o-neck	Petal	natural	microfiber	?	ruffles	animal	1
2	vintage	High	0.0	L	Autumn	o-neck	full	natural	polyester	?	?	print	1
3	Brief	Average	4.6	L	Spring	o-neck	full	natural	silk	chiffon	embroidary	print	2
4	cute	Low	4.5	M	Summer	o-neck	butterfly	natural	chiffonfabric	chiffon	bow	dot	1

Dataset 2

Name: Fashion Product Images Dataset

Source: Kaggle

Format: CSV

Purpose:

- Product information
- Brand analysis
- Product pricing
- Color trends

Reason for Selection:

The dataset contains information about fashion products, including brand, price, gender, product description, and color, which supports market and consumer trend analysis.

4. Data Collection Strategy

The following steps were followed:

1. Identified publicly available datasets from OpenML and Kaggle.
2. Downloaded the datasets.
3. Verified dataset formats.

4. Imported datasets into Python using pandas and scipy.io.arff.
5. Inspected dataset structure and metadata.
6. Renamed columns for readability.
7. Converted byte strings into readable text.
8. Identified missing values and duplicates.
9. Cleaned the data.
10. Saved the cleaned dataset as a CSV file.

Screenshot

	ProductID	ProductName	ProductBrand	Gender	Price (INR)	NumImages	Description	PrimaryColor
0	10017413	DKNY Unisex Black & Grey Printed Medium Trolle...	DKNY	Unisex	11745	7	Black and grey printed medium trolley bag, sec...	Black
1	10016283	EthnoVogue Women Beige & Grey Made to Measure ...	EthnoVogue	Women	5810	7	Beige & Grey made to measure kurta with churid...	Beige
2	10009781	SPYKAR Women Pink Alexa Super Skinny Fit High-...	SPYKAR	Women	899	7	Pink coloured wash 5-pocket high-rise cropped ...	Pink
3	10015921	Raymond Men Blue Self-Design Single-Breasted B...	Raymond	Men	5599	5	Blue self-design bandhgala suitBlue self-desig...	Blue
4	10017833	Parx Men Brown & Off-White Slim Fit Printed Ca...	Parx	Men	759	5	Brown and off-white printed casual shirt, has ...	White

5. Dataset Description

The Dress Recommendation Dataset contains product-related attributes such as:

- Style
- Price Category
- Rating
- Size
- Season
- Neckline
- Sleeve Length
- Waist Line
- Material
- Fabric Type
- Decoration
- Pattern Type
- Recommendation

The Fashion Product Images Dataset contains:

- Product Name
- Brand
- Gender
- Price (INR)
- Color

- Product Description

6. Data Cleaning Methodology

The following techniques were applied:

- Renamed columns
- Decoded byte strings
- Replaced "?" values with missing values (NaN)
- Checked missing values using `isnull()`
- Filled missing categorical values using the mode
- Checked and removed duplicate records
- Standardized data formats
- Saved the cleaned dataset

7. Python Tools and Libraries

- Python
- Google Colab
- Pandas
- NumPy
- SciPy (scipy.io.arff)
- Matplotlib

8. Results

The cleaned dataset:

- Contains meaningful column names
- Has readable text values
- Has missing values handled
- Has duplicate records removed
- Is ready for visualization and analysis

9. Challenges

- Loading ARFF files
- Byte-encoded values
- Missing values represented by "?"
- Multiple categorical attributes
- Dataset standardization

10. Limitations

- Public datasets may contain incomplete records.
- Some attributes have many missing values.
- The dataset may not represent the entire apparel market.
- Consumer preferences may change over time.

11. Conclusion

This project successfully demonstrated the process of collecting and cleaning publicly available apparel datasets. Python and its libraries were used to preprocess the data by renaming columns, decoding byte strings, handling missing values, and removing duplicates. The cleaned dataset is now suitable for further analysis, visualization, and business decision-making.

12. References

1. OpenML – Dress Recommendation Dataset
2. Kaggle – Fashion Product Images Dataset
3. Pandas Documentation
4. NumPy Documentation
5. SciPy Documentation

